FINANCE MANAGEMENT MODULE

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · Product Team · School Finance Department · Administrators

Related Modules:
Student Management · HR & Payroll · Store & Inventory · Procurement · Cafeteria · Assets · Super Admin

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1. MODULE OVERVIEW

1.1 Purpose

The **Finance Management Module** handles all financial operations within the SmartCampus system.

The module manages:

- Student fee billing
- Payment processing
- Expense management
- Budget tracking
- Financial reporting
- Financial compliance
- Revenue analytics
- Institutional accounting

The system is designed to comply with **IPSAS (International Public Sector Accounting Standards)** and standard educational financial practices.

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2. CORE OBJECTIVES

The Finance Module enables schools to:

1. Automate **student fee billing**
2. Track **all payments and transactions**
3. Manage **expenses and procurement costs**
4. Monitor **budgets and financial performance**
5. Generate **financial reports**
6. Ensure **financial transparency**
7. Maintain **audit-ready accounting records**
8. Integrate **payment systems and banking**

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3. SYSTEM ROLES & ACCESS CONTROL

Role	Permissions
Super Admin	Platform-level financial oversight
School Admin	Financial configuration
Finance Manager	Manage financial records
Bursar / Accountant	Process payments and expenses
Auditor	Read-only access
Parents	Pay student fees
Students	View fee balances

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4. MODULE ARCHITECTURE

The Finance Module consists of the following subsystems:

```
Finance Management
│
├── Finance Dashboard
├── Fee Management
├── Student Billing
├── Payment Processing
├── Expense Management
├── Budget Management
├── Financial Ledgers
├── Scholarships & Discounts
├── Refund Management
├── Financial Reports
├── Audit & Compliance
```

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5. FINANCE DASHBOARD

5.1 Overview

The **Finance Dashboard** provides a real-time overview of financial operations.

Dashboard Metrics

- Total Revenue
- Fees Collected
- Outstanding Fees
- Total Expenses
- Net Financial Position
- Scholarships Issued

Example Dashboard

```
-----------------------------------------
Finance Dashboard
-----------------------------------------

Total Revenue: $420,000
Fees Collected: $350,000
Outstanding Fees: $70,000

Expenses: $210,000
Net Balance: $210,000

Scholarships Awarded: $20,000
```

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6. FEE MANAGEMENT

6.1 Overview

The system manages **school fee structures**.

Fee types may include:

- Tuition fees
- Boarding fees
- Library fees
- Transport fees
- Examination fees
- Cafeteria subscriptions

Fee Structure Data Model

```
FeeStructure
- fee_id
- fee_name
- class_level
- academic_term
- amount
- due_date
```

Example:

```
Grade 10 Tuition
Amount: $600
Term: Term 1
Due Date: 15 Feb
```

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7. STUDENT BILLING SYSTEM

The system automatically generates **student invoices**.

Billing Workflow

```
Student Enrollment
      ↓
Fee Structure Applied
      ↓
Invoice Generated
      ↓
Parent Receives Notification
```

Billing Data Model

```
Invoice
- invoice_id
- student_id
- fee_id
- amount
- due_date
- status
```

Invoice status:

- Pending
- Partially Paid
- Paid
- Overdue

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8. PAYMENT PROCESSING

Handles fee collection.

Supported Payment Methods

- Cash
- Bank Transfer
- Mobile Money (M-Pesa)
- Online Payment Gateway
- Cheque
- School Wallet

Payment Workflow

```
Parent makes payment
      ↓
System records transaction
      ↓
Invoice updated
      ↓
Receipt generated
```

Payment Data Model

```
Payment
- payment_id
- student_id
- invoice_id
- amount_paid
- payment_method
- transaction_reference
- payment_date
```

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9. SCHOLARSHIPS & DISCOUNTS

Schools may provide financial assistance.

Types:

- Merit scholarships
- Need-based scholarships
- Staff discounts
- Sibling discounts

Scholarship Data Model

```
Scholarship
- scholarship_id
- student_id
- scholarship_type
- amount
- start_term
- end_term
```

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10. REFUND MANAGEMENT

Handles fee refunds when:

- Student withdraws
- Overpayment occurs
- Scholarship adjustments

Refund Data Model

```
Refund
- refund_id
- student_id
- invoice_id
- amount
- reason
- approval_status
```

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11. EXPENSE MANAGEMENT

Tracks school operational expenses.

Examples:

- Staff salaries
- Utilities
- Procurement
- Maintenance
- Cafeteria supplies

Expense Data Model

```
Expense
- expense_id
- category
- description
- amount
- department
- date
```

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12. BUDGET MANAGEMENT

Schools can define budgets per department.

Example:

```
Science Department
Annual Budget: $50,000
Spent: $28,000
Remaining: $22,000
```

Budget Data Model

```
Budget
- budget_id
- department
- allocated_amount
- spent_amount
- fiscal_year
```

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13. FINANCIAL LEDGERS

Maintains accounting records.

Ledgers

- Student Fee Ledger
- Expense Ledger
- General Ledger
- Budget Ledger

Ledger Data Model

```
LedgerEntry
- entry_id
- account_type
- debit
- credit
- description
- date
```

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14. FINANCIAL REPORTS

Core Reports

- Fee Collection Report
- Outstanding Fees Report
- Expense Report
- Budget Utilization Report
- Scholarship Report

Example Report

```
Fee Collection Report

Term 1 Revenue: $210,000
Paid Students: 820
Outstanding Students: 120
```

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15. ANALYTICS & FORECASTING

Advanced analytics may include:

- Revenue trends
- Cashflow projections
- Budget forecasting
- Payment pattern analysis

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16. PAYMENT REMINDERS & NOTIFICATIONS

Automated alerts for:

- Upcoming fee deadlines
- Overdue invoices
- Payment confirmations
- Scholarship approvals

Notification channels:

- In-app
- Email
- SMS
- Push notifications

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17. SYSTEM INTEGRATIONS

Module	Integration Purpose
Student Management	Student billing
HR & Payroll	Salary payments
Procurement	Purchase expenses
Cafeteria	Meal plan payments
Assets	Capital expenditure
Super Admin	Multi-school financial control

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18. AUDIT TRAIL & COMPLIANCE

All financial actions are logged.

Example:

```
AuditLog
- log_id
- user_id
- action
- transaction_id
- timestamp
```

Example log entry:

```
User: Accountant
Action: Updated Payment Record
Amount: $500
Date: 16 March 2026
```

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19. DATA SECURITY

Security features include:

- Role-based access control
- Encrypted financial records
- Secure payment processing
- Audit logging

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20. SCALABILITY

The system supports:

- Multiple campuses
- Multiple academic terms
- Thousands of students
- High transaction volumes

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21. BULK DATA OPERATIONS

Supports:

- CSV import of invoices
- Excel export of reports
- Bulk payment uploads
- API integrations

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22. FUTURE ENHANCEMENTS

Advanced capabilities:

- AI financial forecasting
- Automated bank reconciliation
- Predictive fee default detection
- Integrated mobile payment apps
- Financial risk analysis